

A phase trap in approximate circuit compilation: how a converged compiler silently breaks interferometric protocols, and a one-gate fix

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Tensor-network circuit compilation (AQC) can dramatically compress the controlled time evolution at the heart of a Hadamard test: on an open XXZ chain we measure a crossover against exact controlled-unitary synthesis at $n = 4$ system qubits, growing to a $36\times$ two-qubit-gate reduction at $n = 7$ and $1514\times$ at $n = 10$, at state infidelity below 3×10^{-4} throughout. We then show that shipping this compression naively destroys the experiment it was built for, with no warning from the compiler: the optimisation objective (state fidelity) is blind to global phase, while a Hadamard test is precisely an interferometer that measures that phase. A compression that converges to state infidelity 3×10^{-4} returns an interference signal χ wrong by 1.22–1.35 on a quantity bounded by $|\chi| \leq 1$. The repair costs one single-qubit gate: the offending phase is available for free at compile time, and a $P(-\theta)$ on the ancilla cancels it exactly, reducing the error to below 0.01. We delimit the hazard precisely: direct observable estimation on the compiled state is provably immune, so the trap is specific to interferometric use. We encountered the same magnitude-hides-phase failure mode twice more in independent settings, including a hardware run whose magnitude-based figure of merit scored a completely decohered circuit as the best performer. Finally, we report a pre-registered test of the compilation advantage on two superconducting processors: it does not survive present-day error rates at $n = 6$, falling short of its own prediction by a factor of twelve, and we quantify the effective error

rate at which it becomes measurable.

1 Introduction

The Hadamard test is the elementary interferometric primitive of quantum computation: an ancilla prepared in superposition controls a unitary U on a system register, and the ancilla statistics estimate the complex overlap $\chi = \langle \psi | U | \psi \rangle$, whose phase carries the physics (for $U = e^{-iHt}$, the spectrum). It underlies quantum phase estimation, Krylov-subspace diagonalisation, and the verification of compiled circuits. Its practical obstacle is the controlled evolution: synthesising an exact controlled- U for an arbitrary n -qubit unitary costs $O(4^{n+1})$ two-qubit gates.

Approximate quantum compiling (AQC) [3, 4] attacks exactly this cost: a classical tensor-network optimiser finds a shallow parameterised circuit whose action on a chosen input state matches the deep target circuit, certified by a state fidelity. It is natural, then, to compile the evolution inside a Hadamard test.

This paper reports three things. First, the compression works, and keeps working well past the sizes where the exact construction is hopeless (Sec. 3). Second — our central observation — the compiler’s own success criterion silently breaks the interferometer it is being compiled for (Sec. 4): state fidelity is a magnitude, a Hadamard test measures a phase, and the gap between those two sentences is an error as large as the signal, invisible to every certificate the compilation stack reports. The fix costs one single-qubit gate. We delimit the hazard by proving — and verifying on the same compiled circuits — that direct observable estimation on the compiled state is immune, so ordinary (non-interferometric) uses of AQC are un-

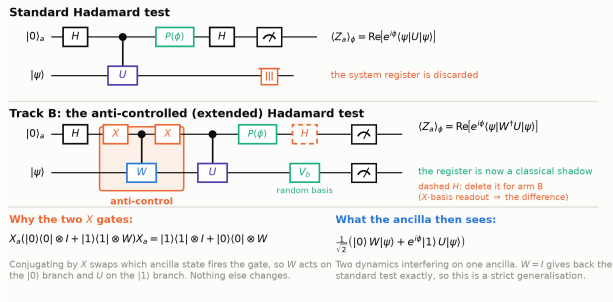


Figure 1: The anti-controlled Hadamard test. Conjugating the controlled- W by X on the ancilla makes W fire on the $|0\rangle$ branch while the ordinary controlled- U fires on the $|1\rangle$ branch, so the ancilla interferes two dynamics: $\langle Z_a \rangle_\varphi = \text{Re}[e^{i\varphi}\chi_{AB}(t)]$ with $\chi_{AB}(t) = \langle\psi|W^\dagger U|\psi\rangle$. Setting $W = I$ recovers the standard test. Construction after Ref. [1], Apps. B and E.

affected (Sec. 5). Third, we practise what the finding preaches: we pre-registered survival predictions for the compiled Hadamard test on two superconducting processors and report their falsification, including a second, independent appearance of the magnitude-hides-phase failure in the hardware data itself (Sec. 6).

The instrument we use to detect and quantify the trap is itself of independent interest: an *anti-controlled* Hadamard test that interferes two dynamics — the compiled circuit W and its target U — on one ancilla, measuring $\chi_{AB}(t) = \langle\psi|W^\dagger U|\psi\rangle$ directly on the device (Sec. 2). The construction follows the appendices of Ref. [1], whose “shadow-enhanced” Hadamard test this project implements end to end; measuring the otherwise-discarded system register in random bases à la classical shadows [2] upgrades the scalar verdict to a per-observable one.

Throughout, the benchmark family is deliberately small (2–10 qubits) and exactly solvable, so that every estimate is graded against exact truth; we claim no advantage over classical computation anywhere in this paper. Every number reported here is reproducible from a public repository whose results ledger records the producing command for each value.

2 The instrument: an anti-controlled Hadamard test

Let $U = e^{-iHt}$ be the target evolution and W any cheaper stand-in — a Trotter product formula or a compiled circuit. Sandwiching a controlled- W

between two X gates on the ancilla makes W act on the $|0\rangle$ branch; an ordinary controlled- U then acts on the $|1\rangle$ branch (Fig. 1). The ancilla expectation values in the two quadratures $\varphi \in \{0, -\pi/2\}$ yield the complex overlap of the two dynamics,

$$\chi_{AB}(t) = \langle\psi|W^\dagger(t)U(t)|\psi\rangle, \quad (1)$$

measured on the device rather than simulated. This is the on-device answer to the question every practical simulation raises: *how wrong is the circuit you can afford, relative to the one you meant to run — and wrong in what?*

Because the circuit is new, it gets its own validation rather than inherited trust: on our benchmark models the statevector identity between Eq. (1) and the constructed circuit holds to 9.0×10^{-15} ($n = 3$) and 4.6×10^{-15} ($n = 2$). Deleting the final Hadamard and measuring the system register in uniformly random Pauli bases (a classical shadow [2]) splits the ancilla-tagged records into estimators of $(W\rho W^\dagger \pm U\rho U^\dagger)/2$, giving every system observable under W and under U *separately* from the same shots — a per-observable disagreement profile on top of the scalar χ_{AB} [1]. We note for completeness that for the scalar overlap alone this instrument is not the cheapest option: a Loschmidt echo is 4–6× cheaper at every size we measured and, on hardware, substantially more accurate; the anti-controlled test earns its keep through the phase and the per-observable profile, which no magnitude-only test returns.

3 Compilation makes the test scale

The benchmark family is an open XXZ chain, $H = \sum_i [J(X_i X_{i+1} + Y_i Y_{i+1}) + \Delta Z_i Z_{i+1}] + \sum_i h_i Z_i$ with $J = 0.65$, $\Delta = 0.25$ and fixed local fields, evolved to $t = 0.9$; the input state is a product state with a single $R_y(1.3)$ rotation. The honest baseline is the direct synthesis of $|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$ as a single $(n+1)$ -qubit unitary — markedly cheaper than controlling a synthesised circuit, and therefore harder to beat.

Compiling the evolution with the MPS-backed AQC optimiser [4, 5] and then controlling the resulting ansatz gives the two-qubit gate counts of Table 1. The crossover sits at $n = 4$; by $n = 7$ the compiled construction is 35.9× cheaper, by $n = 10$, 1514× — while the compiler’s state infidelity

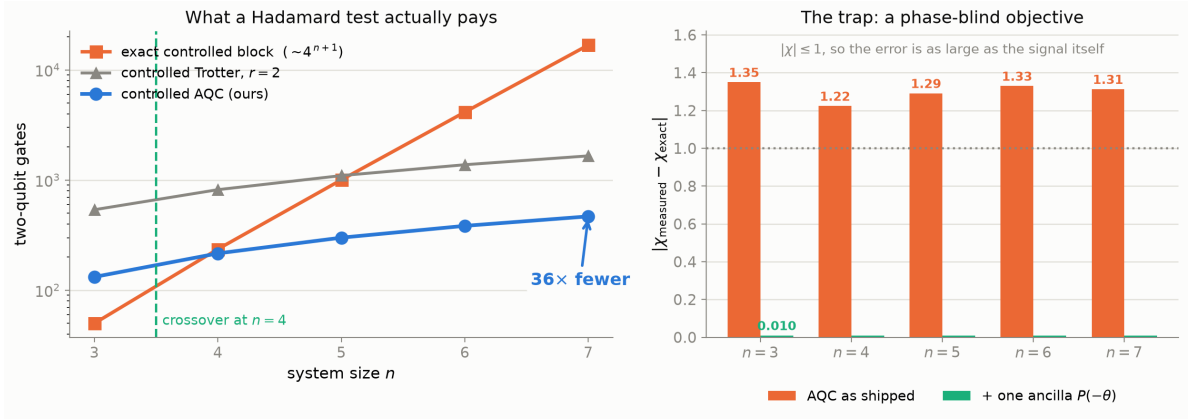


Figure 2: Two-qubit gate cost of the controlled evolution on an open XXZ chain at $t = 0.9$, and the phase trap. The exact controlled block grows as $O(4^{n+1})$; the compiled (AQC) circuit grows roughly linearly, with a measured crossover at $n = 4$. The trap panel shows the compiled circuit’s interference error before and after the one-gate phase correction of Sec. 4.

stays flat at $(1.4\text{--}3.0) \times 10^{-4}$ and the (phase-corrected) interference error stays at 0.008–0.010.

The advantage is not an artefact of the 1D chain or of idealised connectivity. On a 2D lattice the crossover shifts by one qubit and the advantage reaches $100.8\times$ at $n = 8$; on a 2×2 Fermi–Hubbard grid (8 qubits, Jordan–Wigner) the gate advantage is $34.7\times$, though there the compression accuracy degrades by roughly $30\times$ and we do not claim it usable without further optimisation; and re-transpiling every circuit against a real heavy-hex coupling map *widens* the advantage in every case measured (e.g. $10.8\times \rightarrow 15.4\times$ at $n = 6$), because the dense exact block pays about $2.1\times$ for routing while the shallow local ansatz pays about $1.5\times$.

One scope statement belongs in the main text rather than a footnote: the compiled W is optimised for *one* input state, and its process infidelity is accordingly large (~ 0.96 at $n \geq 5$). That is legitimate for a Hadamard test, whose controlled gate only ever acts on that prepared state, and illegitimate the moment anything feeds the compiled gate a different input.

4 The trap: a phase-blind objective breaks a phase interferometer

The AQC objective maximises the state fidelity $|\langle \psi_{\text{target}} | \psi_{\text{ansatz}} \rangle|$ — a magnitude, invariant under $|\psi_{\text{ansatz}}\rangle \rightarrow e^{i\theta} |\psi_{\text{ansatz}}\rangle$. A Hadamard test measures precisely that phase. Concretely, a compression that converges to state infidelity 3×10^{-4}

returns

$$W|\psi\rangle = e^{i\theta} U|\psi\rangle, \quad \theta \text{ up to } 3.0 \text{ rad}, \quad (2)$$

and the measured interference signal comes out wrong by $|\Delta\chi| = 1.22\text{--}1.35$ across $n = 3\text{--}7$ — on a quantity bounded by $|\chi| \leq 1$, an error as large as the signal itself. No certificate the compilation stack reports can see it: the objective, the reported infidelity, and the common “survival” summary $|\chi_{\text{meas}}|/|\chi_{\text{exact}}|$ are all magnitudes.

We nearly missed it ourselves. Our first certification of the compiled circuit used $|\chi_{AB}|$ — also a magnitude — and passed. Only the phase-sensitive readout of the anti-controlled test (Sec. 2) exposed the error, which is the practical argument for phase-sensitive verification of compiled circuits in general.

The fix costs one single-qubit gate. If $W|\psi\rangle = e^{i\theta} U|\psi\rangle$, then $\theta = \arg\langle U\psi | W\psi \rangle$ is available at compile time — the same MPS contraction that certifies the compression evaluates it at negligible extra cost — and a phase gate $P(-\theta)$ on the *ancilla* cancels it exactly, because the ancilla branch carrying W is the one the phase multiplies. The interference error drops from 1.33 to 0.008 at $n = 6$, a $135\text{--}168\times$ improvement across the range, and the correction is included in every “fixed” number of Table 1.

The same magnitude-hides-phase failure appeared twice more, independently. Re-analysing a hardware dataset taken by teammates on a square-lattice device, one run’s survival read 0.988 — near perfect — while the complex χ was 22.6σ from the exact value, a 0.77-rad phase

error; the magnitude survived the device, the phase did not. The third instance appears in our own hardware experiment below (Sec. 6), where a magnitude-only metric crowns a completely dead circuit the best performer.

5 Scope of the trap: pure compilation is immune

It is important not to overclaim: the trap is *not* a defect of approximate compilation itself. In its textbook use — compress a state-preparation circuit, then measure observables directly on the prepared state — AQC is provably unaffected, because an observable expectation value is exactly invariant under the offending phase:

$$\langle O \rangle_W = \langle \psi | W^\dagger O W | \psi \rangle = \langle \psi | U^\dagger e^{-i\theta} O e^{i\theta} U | \psi \rangle = \langle O \rangle_U, \quad (3)$$

for any Hermitian O and any θ . The overlap $\chi = \langle \psi | U | \psi \rangle$, by contrast, is an interference term between two states, not an expectation value in any single state, and extracting it physically requires an ancilla-based circuit — which is exactly where the phase re-enters.

We verified this on the same compiled circuits as Table 1 (the interference errors reproduce to four digits, confirming identical compilations): direct estimation of $\langle Q \rangle$, $\langle H \rangle$ and $\langle Z_0 \rangle$ on the compiled state has worst-case error 0.018 ($n = 3$), 0.014 ($n = 5$) and 0.015 ($n = 7$) — two orders of magnitude below the interferometric error on the same circuit, and consistent with the compiler’s ordinary state infidelity. Multiplying the compiled state by an arbitrary additional global

Table 1: Controlled-evolution two-qubit gate counts (ideal basis-gate counts, no routing) and the compiled circuit’s error measures. State infidelity is the compiler’s own certificate; $|\Delta\chi|$ is the interference error after the one-gate phase correction of Sec. 4.

n	exact	AQC	ratio	state infid.	$ \Delta\chi $ (exact)
3	50	132	$0.38\times$	1.4×10^{-4}	0.0100
4	236	216	$1.09\times$	2.6×10^{-4}	0.0080
5	1004	300	$3.35\times$	2.7×10^{-4}	0.0082
6	4140	384	$10.8\times$	3.0×10^{-4}	0.0079
7	16 812	468	$35.9\times$	2.9×10^{-4}	0.0083
8	67 756	552	$122.8\times$	2.9×10^{-4}	0.0083
9	272 044	636	$427.7\times$	2.9×10^{-4}	0.0083
10	1 090 220	720	$1514\times$	2.8×10^{-4}	0.0082

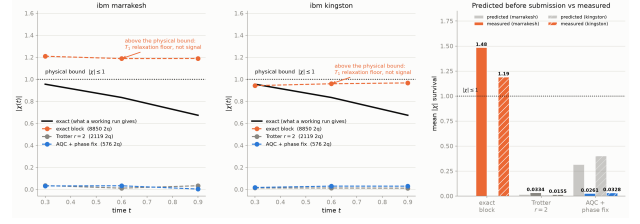


Figure 3: Three constructions of the same $n = 6$ Hadamard test on `ibm_marrakesh` (4 000 shots per circuit, $t \in \{0.3, 0.6, 0.9\}$), against survival predictions recorded before submission. Every arm returned noise; the deliberately hopeless 8850-gate exact arm reports an *unphysical* survival above 1 — a T_1 -relaxation floor that a magnitude-only metric scores as the best result on the plot. The y -axis deliberately extends past $|\chi| = 1$ to show it.

phase changes every measured observable by less than 2×10^{-15} . The trap is therefore specific to embedding compiled circuits inside interferometric protocols — Hadamard tests, phase estimation, overlap and echo-type verification — and it is exactly there that phase-aware certification, or the one-gate correction above, is required.

6 Pre-registered hardware test, and its honest failure

A compilation advantage measured in gate counts is not yet a hardware advantage. We submitted the same $n = 6$ Hadamard test built three ways — exact controlled block (8 850 two-qubit gates after routing), controlled Trotter (2119), and compiled AQC with the phase fix (576) — to two superconducting processors, with survival predictions written down before submission so that failure would be undeniable.

It was. On `ibm_marrakesh` the compiled arm returned mean survival 0.026 against a predicted 0.315; on `ibm_kingston`, 0.033 against 0.401 — twelve times short on both devices, statistically indistinguishable from the 2119-gate Trotter, with mean interference errors ≈ 0.83 on a unit-bounded quantity. The exact arm is the most instructive failure: its “survival” reads 1.49 (marrakesh) and 1.19 (kingston) — unphysical, since $|\chi| \leq 1$. Both quadratures sit pinned near a constant, which is the ancilla relaxed to $|0\rangle$: at 8 850 gates the circuit runs for $\approx 8T_2$, and what the metric scores is T_1 relaxation, not signal. A magnitude-only figure of merit crowns the deadest circuit the winner — the third independent

sighting of the pattern of Sec. 4.

From the measured survivals, the effective error per two-qubit gate is *bounded below* by $\varepsilon_{\text{eff}} \geq -\ln(S)/N_{2q} \approx 6 \times 10^{-3}$ — a bound rather than an estimate, because the measured $|\chi|$ sits near the estimator’s own counting-noise floor $\sigma\sqrt{\pi/2} \approx 0.020$ at these shot counts. A shot-count sweep (2 000/4 000/8 000 shots, same circuits, same device) supports reading this as a genuine error floor rather than shot-starvation: doubling the shots from 4 000 to 8 000 should shrink a pure counting floor by 29%, but the measured survival moved only 7% ($0.026 \rightarrow 0.024$), and the inferred bound stabilises at $6.3\text{--}6.4 \times 10^{-3}$ — with the caveat that the three shot counts are separate jobs on different days, so device drift is not excluded. Visibility of the compiled signal at this depth requires $\varepsilon_{\text{eff}} \lesssim 2 \times 10^{-3}$: a gap of at least $3\times$, roughly one device generation, and not something a larger n improves — transpiling the full experiment at $n = 7$ shows the compilation *ratio* doubling while the absolute compiled depth grows and the predicted signal halves.

7 Discussion

The finding is small to state and broad to apply: *any* pipeline that compiles or optimises circuits against a magnitude-only objective, and then consumes the result interferometrically, inherits an uncertified phase degree of freedom that can be as large as the signal. The instances here — a tensor-network compiler’s objective, a survival summary statistic, and a relaxation floor on hardware — are three faces of one mistake: certifying an interferometer with a magnitude. The cure is equally simple to state: certify with a phase-sensitive quantity (the anti-controlled test of Sec. 2 is one such instrument, and for the scalar question a Loschmidt echo is cheaper), or cancel the phase at compile time, where it is available for free.

Three honest limitations. First, everything here is measured on one benchmark family at 2–10 qubits, exactly solvable by design so that every number is graded against truth; no claim is made beyond it, and none against classical computation. Second, the compiled circuits are single-input-state compressions; their process infidelity is large, and every use beyond the Hadamard test they were compiled for is out of scope. Third,

the hardware negative is two single runs per device configuration with pre-registered predictions, not a device characterisation; the residual factor $\sim 7\times$ between our noise-budgeted prediction and the measured survival is reported as unattributed rather than explained.

Data and code availability

All circuits, evidence files, and the results ledger that sources every number in this paper (row identifiers are embedded as comments in the manuscript source) are public at <https://github.com/mnsh0409/Qiskit-Hackathon-2026>.

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